

OM BHATT

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PROFESSIONAL SUMMARY

Well-rounded student with strong academics and extracurricular experience. Passionate about robotics, particularly autonomous and intelligent systems. Seeking to expand engineering knowledge and experience to design, develop, and optimize mechatronic and robotic systems that advance automation and contribute to the betterment of society.

U.S. Person (No Sponsorship Required)

EDUCATION

Texas A&M University, *B.S - Mechatronics Engineering*
Minor: Embedded Systems

2024 - 2028
College Station, TX

Relevant Coursework: Calculus I–III, Differential Equations, Electricity & Magnetism, Engineering Mechanics, Strength of Materials, Metallic Materials, Fluid Mechanics & Power, Circuit Analysis, Analog Electronics, Digital Electronics, Embedded Systems Development in C, Microcontroller Architecture, Applied Dynamic Systems

Certifications: Simulink Onramp (MathWorks), C++ Programming Foundations, Introduction to Python, FAA Recreational UAS Safety Test (TRUST), Fischer Engineering Design Center (FEDC) Red Badge

Organizations: IEEE TAMU, ASME TAMU, TURTLE Robotics, Texas Aimbots, SOMTECH, TAMUHACK, SASE

RELEVANT EXPERIENCE

Undergraduate Research Assistant

Autonomous Systems Laboratory, Texas A&M University

Aug 2026 - Present
College Station, TX

- Researching and implementing algorithms for autonomous ground robot navigation under [Dr. Sivakumar Rathinam](#) at the Autonomous Systems Laboratory, J. Mike Walker '66 Department of Mechanical Engineering.

R&D Robotics Engineer - Software Subteam

T.U.R.T.L.E Robotics, Disaster Response Observation Network, Texas A&M University

Sep 2025 - Sep 2026
College Station, TX

- Researched and programmed autonomous flight algorithms (`offboard_control_square.2`) using PX4/MAVROS over ROS 2 Humble, validating autonomous flight via 2-drone simulation and 2-3 outdoor test flights for a 10-member team's UAV swarm platform assisting first responders.
- Co-authored and presented the "[Disaster Response Observation Network \(DRON\)](#)" research poster across two university showcases (Nov 2025, Apr 2026), each featuring 21+ advanced projects, detailing autonomous swarm progress and system architecture to an audience of engineering faculty and peers.

Hardware Team Member

Texas Aimbots, Texas A&M University

Sep 2025 - Sep 2026
College Station, TX

- The [Sentry Robot](#) is a fully autonomous ground robot that moves and shoots at enemy robots using computer vision.
- Designed and 3D-printed 4+ ABS structural components in Onshape CAD for Sentry, a fully autonomous ground robot, including a gimbal housing redesigned for a more compact, rigid footprint; installed on the robot.
- Integrated the NVIDIA Jetson Orin, LiDAR, and camera systems to enable real-time perception and autonomous decision-making ahead of the 2026 ARC Championship, an international collegiate robotics competition.

AEOP Summer Robotics Research Intern - LLMs for Safe Navigation REU

Oden Institute - Center for Autonomy - Autonomous Systems Group, UT Austin

June 2025 - Aug 2025
Austin, TX

- Researched autonomous off-road navigation and drift/jump control dynamics under [Dr. Ufuk Topcu](#), [Dr. Christian Ellis](#), and [PhD Student Rwik Rana](#) to architect a low-cost, 1/16 scale autonomous RC car as a modular research platform for drift and jump navigation studies.
- Executed the complete electromechanical assembly by designing and soldering a custom circuit board to power and integrate all components, including a Raspberry Pi 5/Jetson Nano and motor controllers, verified via multimeter testing.
- Programmed a low-level C++/ROS 2 controller for Ackermann steering and throttle control, validating full teleoperation across 5+ test runs, including a successful jump test.
- Ran a visual SLAM algorithm using camera-based perception in the F1Tenth simulator to validate localization and control, laying the groundwork for planned IMU fusion and post-jump re-localization toward fully autonomous operation.

Hatchling Team Member

T.U.R.T.L.E Robotics, Texas A&M University

Jan 2025 - Sep 2025

College Station, TX

- Won 2nd Place out of 30+ competing teams at the [2025 TURTLE Robotics Hatchling Competition](#)
- Designed and CADded key mechanical components in SolidWorks, including a forklift-style lift and single-block gripper mechanism, as part of a 4-person team, engineered for rapid sequential block acquisition.
- Programmed an ESP32 and Arduino to control precise vehicle navigation and operate the multi-stage lift, enabling the robot to retrieve and stack 7 blocks in competition.
- Collaborating with the T.U.R.T.L.E Robotics team at Texas A&M University to enhance robotics expertise and tackle innovative challenges.

Student Researcher: Engineering Science Fairs - Dual-axis Solar Tracker

Science Fair Club, Vista Ridge High School

Sep 2022 - Mar 2024

Cedar Park, TX

- Engineered and developed "[EcoTrack](#)", a novel phototropic solar tracker inspired by nature, which uses thermo-responsive polymers to follow a light source. This design eliminates the need for electrical components in the tracking mechanism, creating a low-cost and low-maintenance solution for developing nations.
- Validated the system through experimentation, demonstrating that EcoTrack harnessed 87 percent more net energy than a traditional dual-axis tracker and 119 percent more than a fixed panel system under identical conditions.
- Achieved [first place](#) across Austin region at Austin Regional Science Fair and advanced to TXSEF State Fair
- Awarded a university tuition scholarship from [Jacobs Engineering](#) (\$2000) and [RICOH Sustainable Development Award](#)
- Elected to lead the role of an officer to help mentor other students on their projects and showcase our school's achievements to attract potential sponsors

Mechatronics Research Intern - WATonoBus: Autonomous Shuttle

Mechatronics Vehicle Systems Lab, University of Waterloo

Jun 2021 - Sep 2021

Waterloo, Canada

- Engineered an external HMI (eHMI): an LED matrix display for [WATonoBus](#), Professor [Amir Khajepour](#)'s autonomous shuttle at [MVS Lab](#), as 1 of 4 team members (2 students, 1 PhD student) on the eHMI subsystem.
- Developed a ROS node parsing intention-based commands over a roserial link, validating all [icon states and display functions](#) in a live outdoor test on the shuttle's campus loop route.

PUBLICATIONS

- I. Wilhite, C. Ambroziak, A. Briggs, R. Kato, **O. Bhatt**, L. Breedlove, V. So, R. Shah, M. Shi, J. Shouba, O. Thomas, C. Weatherspoon, and D. Wheaton, "Disaster Response Observation Network (DRON)," TURTLE Robotics, Apr. 2026. [Conference Poster].

- I. Wilhite, A. Briggs, C. Ambroziak, R. Kato, D. Wheaton, **O. Bhatt**, A. Talal, J. Shouba, C. Xu, O. Thomas, and S. Ahn, "Disaster Response Observation Network (DRON)," TURTLE Robotics, Nov. 2025. [Conference Poster].
- **O. Bhatt**, "Autonomous RC Rally Car for Jump and Drift Racing," under the guidance of Dr. U. Topcu, Dr. C. Ellis, and R. Rana, AEOP Summer Robotics Research REU, UT Austin, Aug. 2025. [Research Presentation & Abstract]

VOLUNTEERING AND LEADERSHIP

Autodesk Ambassador Program

Autodesk, Texas A&M University

Jan 2026 - Present

College Station, TX

- Selected for a competitive ambassador program to represent Autodesk, promoting CAD tools through campus events and workshops.

Texas Science and Engineering Fair Judge

College of Engineering, Texas A&M University

Mar 2025 - Present

College Station, TX

- Evaluate 20+ student research projects annually in the Energy: Sustainable Materials & Designs (2025) and Robotics & Intelligent Machines (2026) categories at the Texas Science and Engineering Fair [TXSEF](#), providing constructive feedback to the next generation of engineers.
- As part of Judge appreciation, George H.W. Bush Combat Development Complex gave TXSEF 2025 judges a tour of the Research Integration Center, which has state-of-the-art technology.

Design Review Officer

T.U.R.T.L.E Robotics, Project Branch, Texas A&M University

Jan 2026 - Aug 2026

College Station, TX

- Evaluated system requirements, CAD/electronics documentation, and BOMs across 20+ projects over 3 Design Review sessions.

Sponsorship Committee Member

T.U.R.T.L.E Robotics, External Branch, Texas A&M University

Sept 2025 - Jan 2026

College Station, TX

- Securing corporate sponsorships for TURTLE, a student-led organization, through a new Corporate Sponsor Outreach Initiative.
- Developing and executing outreach strategies to engage with companies and secure funding for workshops, social events, and advanced technical projects.

Student Council Member

Student Council Club, Vista Ridge High School

Sep 2022 - Sep 2024

Cedar Park, TX

- Contributed over 20 hours of community service by participating and helping organize events including school Christmas decorations, district events, conducting fundraisers, elementary school outreach events and more
- Donated items such as toys and food for charities

SKILLS

Robotics & Autonomy: ROS 2, PX4, MAVROS, SLAM, OpenCV, PWM/PID Control, Teleoperation, Gazebo, RViz, Point Clouds

Hardware and Prototyping: ESP32, Arduino, Raspberry Pi, STM32, FPGA, LiDAR, Cameras, PCB Design, Soldering, Analog Electronics, Oscilloscope & DMM Testing, Data Acquisition Systems

Programming Languages: Python, C/C++, Embedded C, Linux, Shell (Bash/Zsh), $\text{\LaTeX}$, HTML, CSS

Simulation and CAD: SolidWorks, FEA, Autodesk Inventor, Onshape, Bambu Studio, Eagle, Multisim, MATLAB/Simulink

AI & Software Tools: Git/GitHub, Visual Studio Code, Workspace, Microsoft Office, LLM Integration (ChatGPT, Claude, Gemini)

AWARDS

AEOP Summer Robotics Research REU Scholarship (\$6000), Oden Institute Jul 2025
Presented by Army Educational Outreach Program (AEOP) and Oden Institute at the University of Texas at Austin for my summer REU work.

2nd Place - 2025 TURTLE Robotics Hatchling Competition, T.U.R.T.L.E Robotics May 2025
Won 2nd Place at the 2025 TURTLE Robotics Hatchling Competition.

AP Scholar, College Board May 2024
Granted to students who receive scores of 3 or higher on three or more AP Exams.

Vista Ridge High School Scholar: “A” Honor Roll, Vista Ridge High School May 2024
Presented to students who have straight As at the end of the school year in all classes enrolled

Texas Science & Engineering Fair Finalist, TXSEF Mar 2024
Advanced to the Texas Science & Engineering Fair (Statewide Science Fair)

2nd Place at Austin Regional Science Fair, Austin Energy Feb 2024
Placed second in the Energy: Sustainable Materials and Design category across the Austin metropolitan region

Vista Ridge High School Scholar: “A” Honor Roll, Vista Ridge High School May 2023
Presented to students who have straight As at the end of the school year in all classes enrolled

Texas Science & Engineering Fair Finalist, TXSEF Mar 2023
Advanced to the Texas Science & Engineering Fair (Statewide Science Fair)

1st Place at Austin Regional Science Fair, Austin Energy Feb 2023
Placed first in the Energy: Sustainable Materials and Design category across the Austin metropolitan region

Jacobs Engineering Tuition Scholarship (\$2000), Jacobs Engineering Feb 2023
Presented to one selected finalist as a tuition scholarship that can be applied to the college of choice

RICOH Sustainable Development Award, RICOH Feb 2023
Presented to the student whose research has demonstrated the principles and technical innovations that offer the greatest potential for sustainable development

Summer Intern Scholarship, University of Waterloo Jun 2021
Presented by Professor Amir Khajepour to me for my work at Mechatronics Vehicle Systems Lab